SERK Construction OS: work plan

57 of 174 done · 33%

All Open Done | Find a task, WP or SRS id | Expand all | Collapse all

Project start to checkpoint 0.7.0

20 to 26 October 2026: foundations, security gate A, the three business journeys, platform

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SPECIFICATION AND PLANNING

- [x] ~~SRS v1.0 written: 253 requirements, role matrix, test catalogue~~
  6 Nov
  Recorded from the repository status · Nov 8, 2026
- [x] ~~Master Specification v2.1 added to the repository~~
  docs/srs
  Recorded from the repository status · Nov 8, 2026
- [x] ~~Implementation plan: 45 work packages, phases R0 to R4, status board~~
  docs/IMPLEMENTATION_PLAN.md
  Recorded from the repository status · Nov 8, 2026
- [x] ~~UAT plan for release 1~~
  docs/release/UAT_PLAN_R1.md
  Recorded from the repository status · Nov 8, 2026
- [x] ~~UAT manual written from the SRS: 273 cases, 14 scenarios, 253 of 253 requirements traced~~
  docs/uat
  Recorded from the repository status · Nov 8, 2026

FINANCIAL KERNEL AND GATE A (SECURITY AND ACCOUNTING INTEGRITY)

- [x] ~~Financial kernel vertical slice 0.1.0~~
  20 Oct
  Recorded from the repository status · Nov 8, 2026
- [x] ~~Exact-decimal money, ISO 4217 registry, rounding and allocation~~
  pkg/money
  Recorded from the repository status · Nov 8, 2026
- [x] ~~Balanced immutable journals, periods, reversals, integrity scan~~
  SC-01
  Recorded from the repository status · Nov 8, 2026
- [x] ~~FX policies with trusted sources and maker/checker~~
  SC-02, SC-03
  Recorded from the repository status · Nov 8, 2026
- [x] ~~Tenant identity from the database login; row-level security on every table~~
  SC-04
  Recorded from the repository status · Nov 8, 2026
- [x] ~~Tenant-wide scope and project-level access rules~~
  SC-05
  Recorded from the repository status · Nov 8, 2026

- [x] ~~Concurrency of close, post and withdraw~~
  `SC-08`
  Recorded from the repository status · Nov 8, 2026

- [x] ~~Hash-chained audit log; audit and outbox written in one transaction~~
  `SC-07`
  Recorded from the repository status · Nov 8, 2026

- [x] ~~Outbox relay: ordering, back-off, dead-letter, replay, signed webhooks~~
  `internal/relay`
  Recorded from the repository status · Nov 8, 2026

- [x] ~~Earned value with price basis and provenance~~
  `SC-09`
  Recorded from the repository status · Nov 8, 2026

- [x] ~~WBS versions and cost codes with approval~~
  `SC-10`
  Recorded from the repository status · Nov 8, 2026

- [x] ~~Pre-auth abuse limits, shared across API replicas~~
  `SC-13, 0029`
  Recorded from the repository status · Nov 8, 2026

- [x] ~~Fail-closed configuration and production compose (code)~~
  `SC-06`
  Recorded from the repository status · Nov 8, 2026

- [x] ~~OIDC verifier (JWKS, MFA from token claims)~~
  `SC-11 part`
  Recorded from the repository status · Nov 8, 2026

- [ ] go.sum checked against sum.golang.org and govulncheck run
  `SC-12; blocked: unreachable from the dev environment`

APPROVALS AND WORKFLOW

- [x] ~~Workflow engine: governed definitions, quorum, SoD, mandatory database gate, timers~~
  `ADR 0007`
  Recorded from the repository status · Nov 8, 2026

- [x] ~~Twelve independent reviews fixed and regression-tested~~
  `reviews 1 to 12`
  Recorded from the repository status · Nov 8, 2026

- [x] ~~Verified approval provenance: forged approvals refused as tenant SQL and schema owner~~
  `GATEA-APPROVAL-01`
  Recorded from the repository status · Nov 8, 2026

- [x] ~~Isolated approval signer process~~
  `serk-signer`
  Recorded from the repository status · Nov 8, 2026

- [x] ~~PUBLIC EXECUTE revoked; search_path pinned on every function~~
  `GATEA-06`
  Recorded from the repository status · Nov 8, 2026

- [x] ~~Session revocation list checked per request and inside approvals~~
  `GATEA-07 server side`
  Recorded from the repository status · Nov 8, 2026

- [ ] Visual workflow designer
  `WFL-05`

BUSINESS JOURNEYS

- [x] ~~Journey A: BOQ, requisition, RFQ, PO, receipt, invoice, payment, bank reconciliation, ledger~~
  `kernel, API, browser`
  Recorded from the repository status · Nov 8, 2026

- [x] ~~Journey B: contract, measurement, IPC, invoice, receipt, retention release, credit notes, variations~~
  `kernel, API, browser`
  Recorded from the repository status · Nov 8, 2026

- [x] ~~Journey C: stock FIFO and WAC, put-away, issues, counts, transfers, inventory to ledger~~
  `kernel, API`
  Recorded from the repository status · Nov 8, 2026

- [x] ~~Planning: critical path, baselines, progress, earned value~~
  `0027`
  Recorded from the repository status · Nov 8, 2026

- [x] ~~Site diaries, registered devices, idempotent offline sync with conflicts~~
  `0037`
  Recorded from the repository status · Nov 8, 2026

OPERATIONS AND PLATFORM

- [x] ~~Point-in-time restore drill, encrypted and signed off-site copy~~
  `GATEA-09`
  Recorded from the repository status · Nov 8, 2026

- [x] ~~Metrics, trace context, alert rules and an executed alert drill~~
  `GATEA-11 part`
  Recorded from the repository status · Nov 8, 2026

- [x] ~~SBOM, licence review and secret scan (offline part)~~
  `GATEA-12 part`
  Recorded from the repository status · Nov 8, 2026

- [x] ~~Demo staging on Railway with synthetic seed data~~
  `25 Oct`
  Recorded from the repository status · Nov 8, 2026

- [x] ~~Console redesign, PDF/Word/Excel exports, reports hub, notifications~~
  `26 Oct`
  Recorded from the repository status · Nov 8, 2026

- [x] ~~Platform 0038: company portals, automation rules, intelligence, anti-exfiltration~~
  `26 Oct`
  Recorded from the repository status · Nov 8, 2026

- [x] ~~Full automated suite: 378 pass, 0 fail, 0 skip (recorded)~~
  `26 Oct`
  Recorded from the repository status · Nov 8, 2026

- [x] ~~Gate status and NO-GO record generated from evidence~~
  `docs/release`
  Recorded from the repository status · Nov 8, 2026

Phase R0: foundations

Weeks 1 to 6. Exit: CI green on every change; staging with TLS and an identity provider

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WORK PACKAGES

- [x] ~~WP-0.1 Scrollable, collapsible console sidebar~~
  EI-UI-01, NFR-USB-03 · 1 ew · delivered 6 Nov
  Recorded from the repository status · Nov 8, 2026
- [x] ~~WP-0.8 Self-explanatory pages (how this page works)~~
  NFR-USB-02, NFR-USB-06 · 1 ew · delivered 7 Nov
  Recorded from the repository status · Nov 8, 2026
- [x] ~~WP-0.3 Containerised dev environment committed and documented~~
  NFR-MNT-05 · 0.5 ew
  Recorded from the repository status · Nov 8, 2026
- [ ] **WP-0.2 Continuous integration** 2/5
  NFR-SEC-05, NFR-MNT-03 · 2 ew
- [ ] **WP-0.4 Design system and language support** 0/4
  ADM-04, NFR-USB-04, NFR-USB-07 · 5 ew
- [ ] **WP-0.5 Identity provider: OIDC, step-up MFA, revocation** 0/4
  IAM-02, IAM-04, IAM-11 · 4 ew · closes GATEA-02, GATEA-07
- [ ] **WP-0.6 Staging with TLS, vault and KMS-held signer** 0/5
  NFR-SEC-01, NFR-SEC-04, NFR-AVL-05 · 3 ew · closes GATEA-08
  - [ ] Staging host with edge TLS
  - [ ] Secrets served from a vault
  - [ ] Signer key in KMS or HSM after a documented key ceremony
    KEY_CUSTODY_RUNBOOK
  - [ ] Nightly encrypted backup to object-locked storage; restore drill scheduled
  - [ ] Staging passes the production configuration check
- [ ] **WP-0.7 Observability** 0/4
  NFR-MNT-06 · 2 ew · closes GATEA-11

Phase R1: financial core, general availability

Weeks 7 to 20. Exit: gates R1 to R5, R8, R9 passed; penetration test closed; signed GO

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WORK PACKAGES

- [x] ~~WP-1.2 Authority matrix, delegation, duty-conflict analysis~~
  IAM-09, IAM-13, IAM-14 · 4 ew
  Recorded from the repository status · Nov 8, 2026
- [x] ~~WP-1.3 Budget revisions, transfers and contingency; original vs current budget~~

PCC-01 · 5 ew
Recorded from the repository status · Nov 8, 2026

- [x] ~~WP-1.4 Financial control dashboard with drill-down~~
  RPT-03 · 4 ew
  Recorded from the repository status · Nov 8, 2026
- [x] ~~WP-1.5 Report catalogue to 25 and the report framework~~
  RPT-01, RPT-02 · 5 ew
  Recorded from the repository status · Nov 8, 2026
- [x] ~~WP-1.6 Units of measure, reservations and reorder points~~
  INV-04, INV-05, INV-07 · 4 ew
  Recorded from the repository status · Nov 8, 2026
- [x] ~~WP-1.7 Multi-entity structure and project master~~
  ORG-01, ORG-02 · 4 ew
  Recorded from the repository status · Nov 8, 2026
- [x] ~~WP-1.8 Guided onboarding and staged imports~~
  ADM-01, INT-03 · 5 ew
  Recorded from the repository status · Nov 8, 2026

> **WP-1.9 R1 hardening, documentation and release** 3/6
> 4 ew · code-complete, not released

> **WP-1.1 Gate A closure and production readiness** 0/6
> NFR-SEC-03, NFR-SEC-07, NFR-PERF-01..04, NFR-AVL-02 · 6 ew + vendor

Phase R2: project controls and commercial

Weeks 21 to 40. Exit: finance and tax sign-off; payroll fixtures signed by an adviser

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Phase R3: field, documents and QHSE

Weeks 41 to 60. Exit: offline gate R6 passed; field pilot on two live projects

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Phase R4: intelligence and ecosystem

Weeks 61 to 76. Exit: governed-AI gate R7 passed; performance targets met

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Gates, decisions and sign-offs

Nothing here can be ticked by engineering alone. Decision status is not recorded in the repository, so all are open until someone ticks them

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RELEASE GATES (DOCS/RELEASE/GO_NO_GO.MD)

- [ ] Gate A passed: trusted foundation
  not passed: IdP, staging, supply-chain checks outstanding
- [ ] Gate B passed: core ERP

not passed: claims and daywork not built

- [x] ~~Gate C engineering evidence: journeys A and B at kernel, HTTP and browser level~~
  AI-verified
  Recorded from the repository status · Nov 8, 2026
- [ ] Gate C human finance sign-off
- [ ] Gate D passed: field execution
- [ ] Gate E passed: advanced scope
- [ ] Gate F passed: governed AI
- [ ] Gate P passed: production authorisation

OPEN DECISIONS (SRS APPENDIX E)

- [ ] D-01 Mobile: native or PWA
  before R3
- [ ] D-02 Hosting and data residency
  before R1 GA
- [ ] D-03 Identity provider and signing-key custody
  before R1 GA
- [ ] D-04 Payroll statutory rules and adviser
  before R2
- [ ] D-05 IFRS 15 method, WIP policy, JV accounting
  before R2
- [ ] D-06 Tax-authority e-invoicing format
  before R2
- [ ] D-07 Contract templates in scope
  before R2
- [ ] D-08 Bank and mobile-money integrations for release 1
  before R2
- [ ] D-09 Retention periods and legal holds
  before R1 GA
- [ ] D-10 External AI model policy
  before R4
- [ ] D-11 Licensing model
  before R1 GA
- [ ] D-12 Languages at launch
  before R1 GA

HUMAN SIGN-OFFS AND PROFESSIONAL OPINIONS

- [ ] Engineering lead signs the GO record
- [ ] Security lead signs
- [ ] Operations / SRE signs

- [ ] Finance and accounting policy owner signs
- [ ] Product owner signs
- [ ] Tax adviser signs the VAT and withholding mapping
  `NFR-PRV-05`
- [ ] Payroll adviser signs the statutory fixture
  `D-04`
- [ ] Auditor or adviser signs the IFRS 15 and WIP policy
  `D-05`
- [ ] Counsel signs the data-protection and retention mapping
  `NFR-PRV-01, NFR-PRV-03`

USER ACCEPTANCE TESTING (DOCS/UAT)

- [ ] UAT release R1 run and signed
  `T-IAM, T-PRC, T-INV, T-CON, T-FIN, T-TRS, T-WFL, T-AUD, T-RPT, T-ADM`
- [ ] UAT release R2 run and signed
- [ ] UAT release R3 run and signed; field pilot
- [ ] UAT release R4 run and signed
- [ ] Non-functional evidence collected for each release
  `chapter 31`

HOUSEKEEPING

- [ ] Commit the R1 work and forecasting (WP-0.1 to 1.9, 2.1, docs/uat) on branch new-version
  `currently uncommitted`
- [ ] Run the browser journeys in CI and add a sidebar screenshot to the evidence
  `left over from WP-0.1`
- [ ] Refresh seed data and testmanual.md for every new work package
  `definition of done`